Reference Notes

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In LaTeX, the rule is strict: you must use `\newcommand` to create a macro from scratch, and `\renewcommand` to overwrite a macro that already exists (like `\maketitle`).

If you use the wrong one, the compiler will throw a fatal error to prevent you from accidentally breaking core LaTeX functions.

1. Creating a Brand New Command (`\newcommand`)

Use this when defining a custom shortcut, formatting style, or text block that does not exist in standard LaTeX.

Simple Shortcut (No Arguments):

Welcome to Acme Corporation.

Dynamic Command (With Arguments): You can pass up to 9 variables by specifying the number of arguments in square brackets `[1]`, and referencing them inside the definition with `#1`, `#2`, etc.

Please **do not** press the button.

Multi-Argument Command:

Q: What is the time?
A: It is 5 PM.

2. The Three Definition Behaviors

To prevent compilation crashes when sharing templates, LaTeX provides three specific ways to declare commands:

Command	Behavior	When it Throws an Error
`\newcommand`	Creates a completely new macro.	If the name is already taken (e.g., trying to name it `\section`).
`\renewcommand`	Overwrites an existing macro.	If the name does *not* exist yet (LaTeX won't let you renew nothing).

Command	Behavior	When it Throws an Error
`\providecommand`	Creates a macro *only* if it doesn't exist.	Never. If the command already exists, it silently ignores your new code and keeps the old one.